

Tetrahydromethanopterin as a Two-Carbon Carrier: Formation of N^5 -Ethyl- and N^5,N^{10} -Ethylene-Tetrahydromethanopterin in *Methanothermobacter marburgensis*

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ABSTRACT: Tetrahydromethanopterin (H_4MPT) is the main carbon carrier in methane metabolism, which allows the interconversion of a formyl group into a methyl group. Although typically described as a single carbon (C1) carrier, we identified and characterized N^5 -ethyltetrahydromethanopterin (N^5 -ethyl- H_4MPT) and N^5,N^{10} -(1,1-ethylene)tetrahydromethanopterin (N^5,N^{10} -ethylene- H_4MPT) in the model methanogen *Methanothermobacter marburgensis*. The chemical competence of H_4MPT to accept C2 moieties was assayed through the study of the spontaneous formation of N^5,N^{10} -ethylene- H_4MPT by the condensation of acetaldehyde with H_4MPT under standard conditions. The rate constant for this reaction was determined to be $1.53 \pm 0.05 \text{ M}^{-1} \text{ s}^{-1}$, and the equilibrium constant was determined to be $(8.8 \pm 0.5) \times 10^3 \text{ M}^{-1}$, which is ca. 35 times higher compared to the analogous reaction with the more common carbon carrier tetrahydrofolate. Biochemical assays with *M. marburgensis* cell lysate suggest that the observed formation of N^5 -ethyl- H_4MPT relies on the enzymatic reduction of N^5,N^{10} -ethylene- H_4MPT . These findings illustrate the chemical competency of H_4MPT to promote biocatalysis with two-carbon moieties and highlight that such reactions might be compatible with established H_4MPT -mediated microbial pathways.

Tetrahydromethanopterin (H_4MPT) (Figure 1A) is a metabolic one-carbon carrier analog of the more widely utilized tetrahydrofolate (H_4F) (Figure 1B).^{1,2} H_4MPT and H_4F enable enzyme-catalyzed oxidoreductions of the carried carbon between the oxidation states of the formyl and methyl group. Both carriers can bind single carbon units at reactive nitrogen atom 5 (N^5) that is in conjugation with the pterin ring system and involve the nitrogen atom of the arylamine (N^{10}) to form the cyclized intermediate species N^5,N^{10} -methylenyl and N^5,N^{10} -methenyl. H_4MPT , however, differs from H_4F by having a more reactive N^{10} due to the absence of a carbonyl group that is conjugated *para* to the aromatic system (protonated aniline N^{10} : $pK_a = +2.4$ for H_4MPT , vs $pK_a = -1.2$ for H_4F).¹ This higher reactivity contributes to an enhanced stability of the N^5,N^{10} -bound H_4MPT forms, thus facilitating the cyclization of N^5 -formyl to N^5,N^{10} -methenyl and the oxidation of N^5 -methyl to N^5,N^{10} -methylene groups.

H_4MPT is crucial for methane formation in methanogenic archaea and for several metabolic pathways that involve the oxidation of single carbon units to CO_2 (Figure 1C).^{2,3} Hydrogenotrophic methanogens reduce CO_2 to methane with H_2 -derived electrons via formyl-, methenyl-, methylene- and methyl-units bonded to H_4MPT .³ Anaerobic oxidation of methane occurs via the same metabolic steps as in hydrogenotrophic methanogenesis but in the reverse direction.⁴ Alternative methanogenic pathways (e.g. methylotrophic, acetoclastic, or methoxytrophic) can also generate CO_2 by oxidizing N^5 -methyl- H_4MPT , following the same metabolic cascade.^{3,5} In aerobic methane oxidation, methane is first oxidized to methanol and then to formaldehyde that condenses with H_4MPT to N^5,N^{10} -methylenyl- H_4MPT for further

oxidation steps.^{6,7} In archaeal multicarbon alkane oxidation, the carbon chains are proposed to be split via an unknown pathway into acetyl-CoA units, from which C1 moieties are converted to N^5 -methyl- H_4MPT and oxidized analogously.^{8,9}

Despite H_4MPT being utilized for diverse metabolisms, this cofactor has been reported exclusively to be a one-carbon carrier. Here, we identified N^5 -ethyl- H_4MPT and N^5,N^{10} -ethylene- H_4MPT in the model hydrogenotrophic methanogen *Methanothermobacter marburgensis*. To isolate free H_4MPT (with no carbon unit bonded to the N^5 or N^{10}) from biomass, the CO_2/H_2 growth gas mix was switched to pure H_2 prior to harvesting to promote the reduction of loaded H_4MPT (with a carbon bonded to the N^5 and/or N^{10}) toward the release of methane. Unexpectedly, the isolated H_4MPT pool contained a second pterin species in ratios up to ca. 1:1 relative to free H_4MPT . $^1H/^{13}C$ HSQC and HMBC NMR experiments allowed us to identify this additional species to be N^5 -ethyl- H_4MPT , with the ethyl group bonded to N^5 (Figure 2). This species is a homologue of the central intermediate N^5 -methyl- H_4MPT that contains an ethyl instead of a methyl group, which prompted us to investigate whether other C2 homologues of H_4MPT could also be formed by this microbe. Notably, N^5,N^{10} -ethylene- H_4MPT (homologue with a 1,1-ethylene group) was detected

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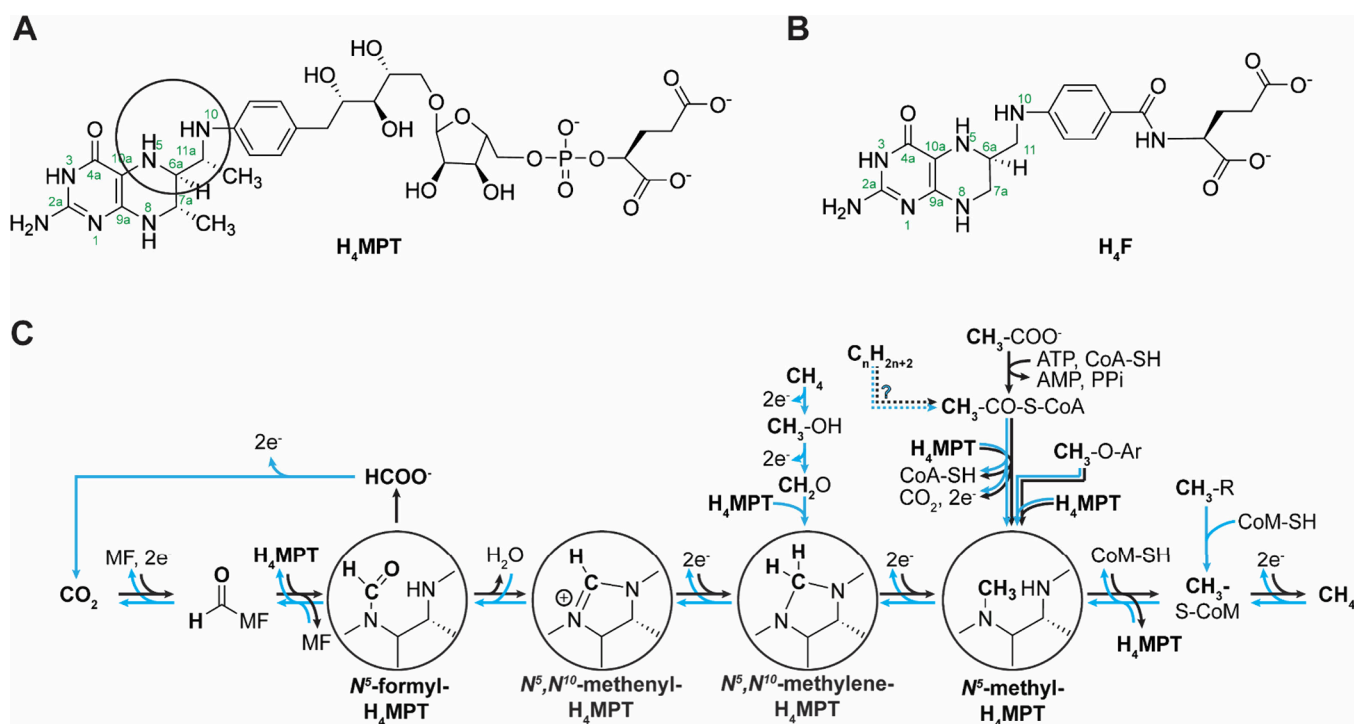


Figure 1. (A) Structure of tetrahydromethanopterin (H_4MPT) with reactive N^5 and N^{10} atoms circled. (B) Structure of tetrahydrofolate (H_4F). (C) Nonexhaustive overview of reported methane-generating and/or CO_2 -generating H_4MPT -mediated carbon redox catalytic cascades in reductive (black arrows) or oxidative (blue arrows) directions. R = $-OH$, $-SH$, $-SCH_3$, $-NH_2$, $-NHCH_3$, and $-N(CH_3)_2$; MF, methanofuran; CoM, coenzyme M; CoA, coenzyme A; ATP, adenosine triphosphate; AMP, adenosine monophosphate; PPI, pyrophosphate; Ar, aromatic residue.

when the cells were harvested under a conventional CO_2/H_2 growth gas mixture (Figure S1).

To confirm the identity of N^5,N^{10} -ethylene- H_4MPT and N^5 -ethyl- H_4MPT , both species were synthesized from free H_4MPT . N^5,N^{10} -Ethylene- H_4MPT is expected to be accessible via acetaldehyde condensation with H_4MPT , analogously to the reported synthesis of N^5,N^{10} -ethylene- H_4F from H_4F .¹⁰ The condensation of purified H_4MPT with different acetaldehyde concentrations was followed via UV spectroscopy under standard conditions (pH = 7.0 and $T = 25^\circ C$) and the dissociation of N^5,N^{10} -ethylene- H_4MPT to acetaldehyde and free H_4MPT via *in situ* 1H NMR spectroscopy (Figure S2). The kinetic parameters of the reaction (Table 1) were quantified by fitting time-dependent concentration data of the individual species to a reaction model. Considering free acetaldehyde (and not the sum with hydrated forms), the second-order kinetic constant for the condensation ($k_{C,obs}$) and the first-order kinetic constant for the hydrolysis ($k_{H,obs}$) were determined as $k_{H,obs} = (1.75 \pm 0.11) \times 10^{-4} s^{-1}$ and $k_{C,obs} = 1.53 \pm 0.05 M^{-1} s^{-1}$, respectively. The hydrolysis reaction monitoring via 1H NMR spectroscopy allowed simultaneous quantification of acetaldehyde and hydrated acetaldehyde along with free H_4MPT and N^5,N^{10} -ethylene- H_4MPT . Therefore, the equilibrium constant of the reaction was determined to be $K_{obs} = (8.8 \pm 0.5) \times 10^3 M^{-1}$. This value is approximately 35-fold higher than the reported value for the analogous reaction with H_4F ¹⁰ ($K_{obs} = 249 M^{-1}$, calculated from the published value of $K_{obs} = 121 \pm 2 M^{-1}$ that is based on total acetaldehyde concentration, considering the hydration constant of acetaldehyde,¹¹ $K_{hydration} = 1.06$, and neglecting acetaldehyde oligomers). The stronger acetaldehyde-binding capabilities of H_4MPT compared to H_4F are consistent with the higher reactivity of its N^{10} that stabilizes the condensed form,¹ making H_4MPT a more suitable cofactor for the

processing of an ethylene group (experimental details in Supporting Information, section II-14 to II-16).

N^5 -Ethyl- H_4MPT was synthesized by potassium borohydride reduction of N^5,N^{10} -ethylene- H_4MPT , analogous to the reported method for the synthesis of C1 H_4MPT species.¹² Both species, alongside their C1 homologues, were characterized via UV and 1H NMR spectroscopy at pH 7.0. The resulting library of H_4MPT species allows their quantification in complex mixtures by integrating the 1H NMR signals from their aniline rings (2b/6b and 3b/5b) and those from the structural methyl groups of their pterin moiety (12a and 13a) (Figure 3).

To study the requirements for the formation of N^5 -ethyl- H_4MPT in *M. marburgensis*, we carried out *in vitro* experiments with lysates of *M. marburgensis* cells that were harvested under the conventional CO_2/H_2 growth gas mixture from bioreactor cultures. The cell lysates were sparged with either pure H_2 or pure N_2 , and the pterin pool was isolated. 1H NMR analysis of the resulting fractions shows that H_2 treatment led to >75% N^5 -ethyl- H_4MPT , whereas untreated and N_2 -treated cell lysate extracts were found to contain less than 10% N^5 -ethyl- H_4MPT (Figure 4A). These *in vitro* assays align with the high level of N^5 -ethyl- H_4MPT that was observed *in vivo* when cells were treated with H_2 in the absence of CO_2 before harvest. The hydrogen requirement for both *in vivo* and *in vitro* formation of N^5 -ethyl- H_4MPT suggests that the ethylation of H_4MPT requires H_2 -derived reducing equivalents in addition to an endogenous two-carbon donor.

To increase the fraction of H_4MPT species that contain a two-carbon moiety in our *in vitro* cell lysate experiments, acetaldehyde was added as a two-carbon donor after gas treatment. The combination of acetaldehyde and H_2 treatment resulted in N^5 -ethyl- H_4MPT being the most abundant species in the isolated pterin pools (Figure 4B). Adding acetaldehyde in

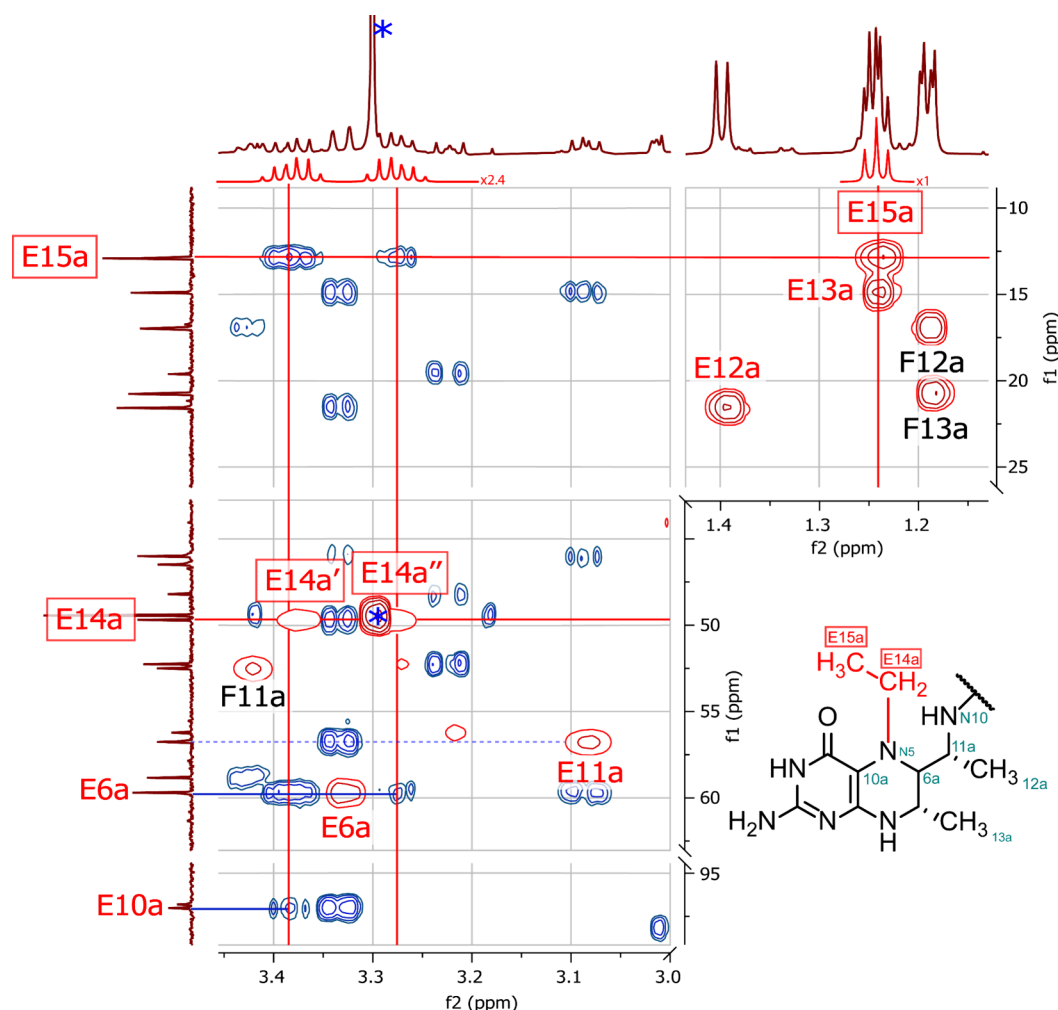


Figure 2. Overlay of 600 MHz $^1\text{H}/^{13}\text{C}$ -2D-NMR spectra ($f1 = ^{13}\text{C}$, $f2 = ^1\text{H}$) of a ca. 1:1 mixture of N^5 -ethyl- H_4MPT and free H_4MPT . Red contour plot: HSQC. Blue contour plot: HMBC. The ^1H NMR spectrum of the N^5 -ethyl group is simulated (red horizontal trace) to help identify the signals of the CH_3CH_2 group that overlap with the methyl groups on the pterins. The signals of the diastereomeric protons of the CH_3CH_2 group overlap with E6a and with remaining methanol (labeled with *). HSQC signals from N^5 -ethyl- H_4MPT are labeled with red E, and those from free H_4MPT , with black F. The identified characteristic signals of the N^5 -ethyl group (boxed red labels, red solid lines) originate from the diastereotopic protons of the methylene group ($-\text{CH}_2\text{CH}_3$, E14a' and E14a'') and from the terminal methyl group ($-\text{CH}_2\text{CH}_3$, E15a). The bonding of the ethyl group to the N^5 is deduced from the HMBC cross-peaks (solid blue lines) of both protons of the CH_3CH_2 group with carbon 6 ($^3J_{\text{C,H}}$) and of the downfield proton of the CH_3CH_2 group with carbon 10 ($^3J_{\text{C,H}}$). No cross-peaks between the signals of the CH_3CH_2 group to those of carbon 11 are observed, which would be a $^4J_{\text{C,H}}$ (blue dotted line). The ethyl group cannot be bonded to N^{10} , because the observed cross-peak between the signals of the CH_3CH_2 group and those of carbon 10 would be a visible $^6J_{\text{C,H}}$ coupling, while a $^3J_{\text{C,H}}$ coupling to carbon 11 is absent (see the Supporting Information, section I-1).

Table 1. Kinetic and Thermodynamic Constants for Acetaldehyde Condensation with H_4MPT

Constant		Value $\pm$ CI 95%
Condensation rate constant	$k_{\text{C,obs}}$	$1.53 \pm 0.05 \text{ M}^{-1}\text{s}^{-1}$
Hydrolysis rate constant	$k_{\text{H,obs}}$	$(1.75 \pm 0.11) \times 10^{-4} \text{ s}^{-1}$
Equilibrium constant	$K_{\text{C,obs}}$	$(8.8 \pm 0.5) \times 10^3 \text{ M}^{-1}$
Gibbs free energy	$\Delta G^{\circ}/^\circ\text{C}$	$-22.51 \pm 0.13 \text{ kJ}\cdot\text{mol}^{-1}$

N_2 -gassed extracts also increased the proportion of isolated N^5 -ethyl- H_4MPT , suggesting that the cells contain reducing equivalents for the *in vivo* formation of N^5 -ethyl- H_4MPT even without additional H_2 treatment. Conversion of acetaldehyde to the ethyl group of N^5 -ethyl- H_4MPT was confirmed by analogous cell lysate experiments with $^{13}\text{C}_2$ -acetaldehyde and H_2 treatment, where the resulting isolated pterin pools showed comparable proportions of N^5 -ethyl- H_4MPT but with a $^{13}\text{C}_2$ -labeled ethyl group, as confirmed via HSQC NMR spectroscopy

(Figure S7). The requirement of a cell lysate for this reaction was demonstrated by the absence of spontaneous formation of N^5 -ethyl- H_4MPT from acetaldehyde and free H_4MPT under a H_2 atmosphere (Figure 4C).

Our results highlight that (1) cell-lysate-promoted formation of N^5 -ethyl- H_4MPT relies on reducing equivalents, (2) acetaldehyde can act as a C2 moiety donor in this reaction, and (3) acetaldehyde can condense with free H_4MPT to form N^5,N^{10} -ethylene- H_4MPT under physiological conditions. These three findings support the hypothesis that N^5 -ethyl- H_4MPT is formed by the reduction of N^5,N^{10} -ethylene- H_4MPT . This reaction would be analogous to the metabolic step catalyzed by the enzyme methylene- H_4MPT reductase (Mer) but proceeding with C2 units on H_4MPT instead of C1 units (Figure 4D). The Mer reaction is known to acquire the reducing equivalents ultimately from hydrogen in hydrogenotrophic methanogens. To confirm that N^5 -ethyl- H_4MPT formation is catalyzed by a

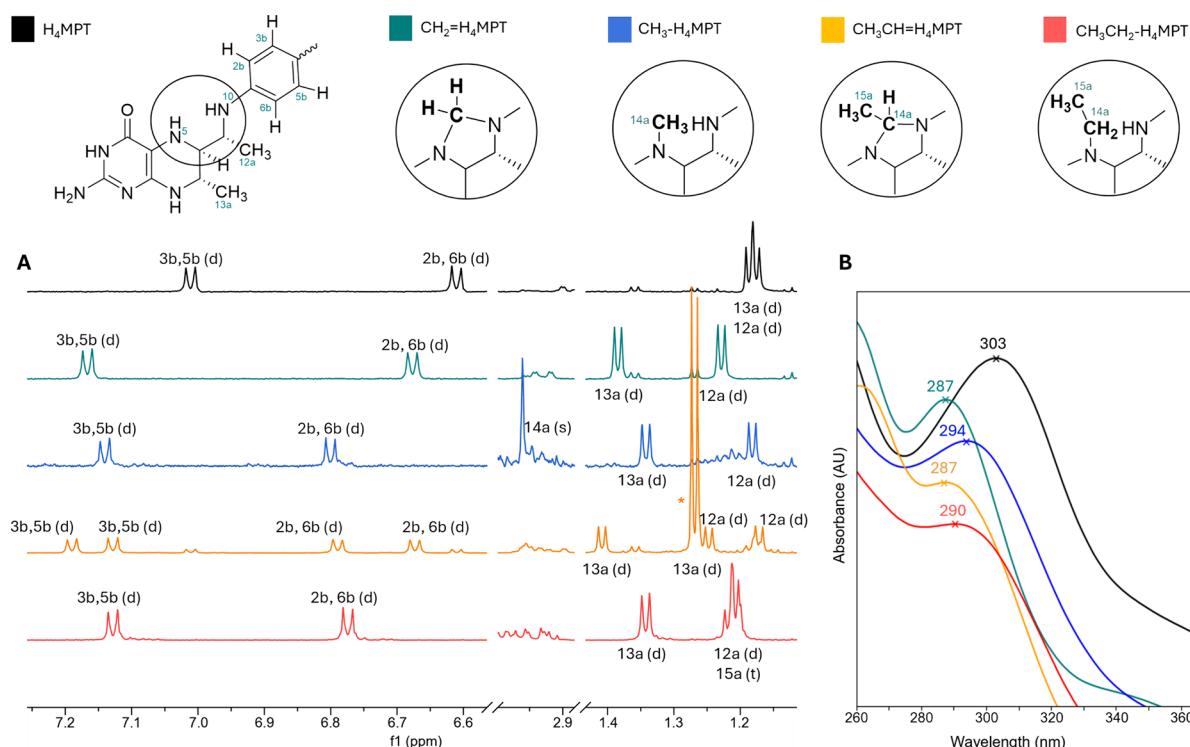


Figure 3. 600 MHz 1H NMR and UV spectroscopic characterization of H_4MPT (black) and its methylene (teal), methyl (blue), ethylene (orange), and ethyl (red) derivatives at pH = 7.0. (A) Characteristic 1H NMR signals of aromatic protons (3b/5b, 2b/6b) and structural methyl groups (12a/13a). To distinguish between the methyl and ethyl derivatives, the 14a-methyl signal at 2.96 ppm is characteristic of N^5 -methyl- H_4MPT , and the 15a- CH_3CH_2 signal at 1.21 ppm that overlaps with 12a is characteristic of N^5 -ethyl- H_4MPT (see the Supporting Information, section I-1 and Figure S3). For N^5,N^{10} -ethylene- H_4MPT , both diastereoisomers (stereochemistry at $CH_3CH=$ of the ethylene group) are present in a ratio of ca. 50:50. $CH_3CH=$ groups are located at 0.89 and 1.47 ppm. One $CH_3CH=$ group is at 5.41 ppm, and the other one overlaps with the water signal (see the Supporting Information, section I-2 and Figure S4). *Excess acetaldehyde hydrate utilized to generate N^5,N^{10} -ethylene- H_4MPT overlaps with 13a. (B) UV profiles of each H_4MPT species and their λ_{max} values. The measured absorbances at λ_{max} range from 0.4 to 0.9 AU (Figure S5). The spectra were normalized at their respective λ_{max} value to free H_4MPT and staggered by -0.05 AU intervals for better visualization.

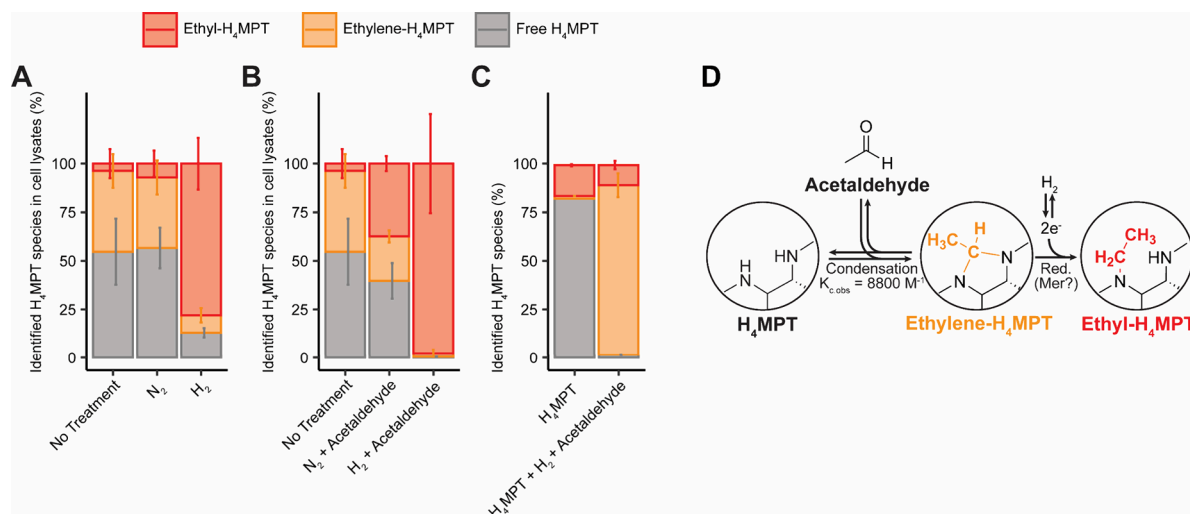


Figure 4. Formation of N^5 -ethyl- H_4MPT and N^5,N^{10} -ethylene- H_4MPT in cell lysates of *M. marburgensis*. (A) Influence of H_2 on the H_4MPT pool. The relative fractions of H_4MPT , ethyl- H_4MPT and ethylene- H_4MPT in the purified pterin pool were quantified via 1H NMR spectroscopy after different treatments. No treatment corresponds to a 3 s flushing with N_2 . Treatments included 90 min of flushing with either N_2 or H_2 . Prior to purification, all samples were incubated for 1.5 h at $60^\circ C$, which is close to the optimal growth temperature for *M. marburgensis*. (B) Same assays but with the addition of 3 mM acetaldehyde after gassing in order to maximize the amount of *in situ*-produced N^5,N^{10} -ethylene- H_4MPT . (C) Cell lysate-free assay: extracted H_4MPT containing minor N^5 -ethyl- H_4MPT vs reaction mixture 25 mM acetaldehyde, 0.25 mM H_4MPT , 100% H_2 . Inter- and intraexperiment statistical comparative *p*-values are available in Figure S6. (D) Putative *in vitro* reaction cascade between acetaldehyde and N^5 -ethyl- H_4MPT .

substrate-promiscuous Mer, additional experiments with the purified enzyme would be required.

The two-carbon units of N^5 -ethyl- H_4MPT and N^5,N^{10} -ethylene- H_4MPT detected in cell-lysates of *M. marburgensis*

are likely formed through carbon salvage pathways and accumulate under our bioreactor conditions as dead-end metabolites. Acetaldehyde produced by the oxidative degradation of H_4 MPT itself in methanogens¹³ could be a potential source of C2 moieties. Endogenous acetaldehyde could also be produced from the decarboxylation of cellular pyruvate, which has been described in the anaerobic archaeon *Pyrococcus furiosus* to be catalyzed as a side-reaction of the enzyme pyruvate:ferredoxin oxidoreductase (Por), particularly under CO_2 limiting conditions.¹⁴ Por is common in anaerobic microbes,¹⁵ and a homologue is also present in *M. marburgensis*.¹⁶ The identification of endogenous two-carbon donors with our experimental approach would require a catalytically active cell lysate without two-carbon donors.

In conclusion, we show that H_4 MPT spontaneously condenses with acetaldehyde to a C2-loaded form that is ca. 35 times more stable compared to the H_4 F analog and that C2- H_4 MPT derivatives can be formed via intracellular processes. These findings illustrate not only the chemical competency of H_4 MPT for two-carbon reactions but also the putative compatibility of such reactions with established H_4 MPT-mediated microbial pathways. Consequently, the pool of metabolites and enzymatic reactions to be considered in biological systems needs to be extended from established C1 units to C2 units. Derivatives of H_4 MPT that carry multicarbon units may serve as substrates for synthetic biocatalytic applications, and they may have already been utilized by nature for the catalysis of non-C1 H_4 MPT-mediated redox reactions. It is tempting to speculate that the unknown catabolic steps in archaeal ethane oxidation⁸ rely on the reaction cascade from N^5 -ethyl- H_4 MPT to acetaldehyde to reroute an ethyl group into established pathways.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.biochem.6c00178>.

Method section, additional data for UV characterization of H_4 MPT derivatives, plots of kinetic model fitting to experimental data for condensation and hydrolysis experiments, cell lysate experiments statistic data, NMR and MS characterization data for free H_4 MPT/ N^5 -ethyl- H_4 MPT mixture and N^5,N^{10} -ethylene- H_4 MPT (PDF)

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*The manuscript was written through contributions of all authors. M.G.L. and H.T. contributed equally to this work.

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Notes

The authors declare no competing financial interest.

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